

Abstract

Repeated cross-sectional (RCS) surveys are widely employed by government agencies and organizations to monitor and model evolving metrics. When conducted at high frequencies, these surveys are prone to be temporally correlated. Existing methods for modeling data of this type either rely on time-step aggregates that result in information loss or fail to properly account for variability when estimating model parameters. In response to this gap in existing methods, we have developed REMARCS (REgression Models for Autocorrelated Repeated Cross-Sections). The accompanying method employs an iterated GLS estimator that fits residuals with an ARMA-Noise model generalized to accommodate variable sample sizes. The efficacy of this algorithm is demonstrated both with simulated data and a case study involving household food security survey data collected by the World Food Programme.

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